Ayush Agarwal

asagarwal04@gmail.com | +91 8451928669 | LinkedIn | GitHub | Portfolio

Summary

Data Engineer specializing in building scalable, automated data infrastructure and robust ETL pipelines. Adept at leveraging PySpark, advanced SQL, and modern cloud data warehouses (Snowflake, Databricks) to optimize large-scale data extraction. Proven ability to clean, transform, and ensure the high availability of complex datasets to feed downstream analytical models and enterprise dashboards.

Skills

Languages & Databases: Python, Advanced SQL (Window Functions, CTEs)
Data & Processing: pandas, Numpy, PySpark, Databricks, Snowflake
Tools & Cloud: Docker, AWS, Git/GitHub, VS Code
Familiarity / Ongoing Learning: Apache Airflow, Google Cloud Platform (BigQuery)
Core Competencies: ETL Pipelines, Cloud Data Warehousing, Data Architecture, System Optimization

Certifications & Training

- **Snowflake Data Cloud:** Completed foundational technical pathways and earned official data warehousing learner badges.
- **Artificial Intelligence using TensorFlow:** SmartInternz & Google.

Experience

Labcorp Bengaluru, India
Data Analyst Associate Intern Feb 2026 – Aug 2026

- Architected and optimized robust ETL pipelines using Python and SQL to extract, clean, and load large-scale datasets into centralized reporting environments.
- Automated manual data transformation processes, ensuring high data availability for enterprise dashboards and accelerating pipeline execution time.
- **Tools Used:** PySpark, Python, Advanced SQL, Snowflake, Databricks

Core Projects

QuickBite Express – High-Performance Optimization Engine *Python, Streamlit, Scikit-learn*

- Engineered an intelligent operational optimization engine utilizing machine learning to streamline delivery dispatching and routing decisions.
- Implemented predictive models to accurately estimate demand surges and dynamic constraints, improving overall processing efficiency.

InsightIQ – Automated Data Ingestion & Schema Mapping Engine *Python, SQL, PostgreSQL, Git*

- Built an automated metadata extraction framework that maps physical database schemas into structured system contexts for processing.
- Optimized custom query-execution validation pipelines to handle structural exceptions and prevent corrupted transactions across downstream environments.

NYC Rental Arbitrage Engine *Python, SQL, ETL Pipelines, XGBoost*

- Engineered a robust ETL pipeline to extract, clean, and integrate complex real estate market datasets for downstream predictive modeling.
- Automated data transformation workflows, ensuring high data availability and structural integrity for interactive financial dashboards.

Research & Publications

Predictive Modelling of Physicochemical Properties (QSPR & QSAR) *Letters in Applied NanoBioScience*

- Authored multiple research manuscripts focused on predicting chemical bioactivity using Topological Indices, deploying ensemble regression models (Random Forest, AdaBoost) to optimize computational accuracy.
- **Published QSPR Paper:** [Link to Paper / DOI: 10.33263/LIANBS143.192]

Urban Plant Suitability Index (UPSI) Framework *Environmental and Sustainability Indicators*

- Designed and computed a novel multi-criteria decision-making framework to evaluate tropical urban ecosystems by synthesizing diverse environmental variables into a single statistical scoring metric.
- **Published Paper:** [Link to Paper]

Education

Your University Name Bengaluru, India
Bachelor of Technology (B.Tech.) in Your Engineering Branch Aug 2022 – June 2026
Cumulative GPA: **8.69/10.0**